Padraig Lavery

Software Engineer – Node.js APIs for Regulated Financial Services

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and running Node.js and .NET services for UK and Middle East banking clients, in a regulated domain where audit trails, input validation and security fixes matter. Backend-leaning, with full stack feature work across web and mobile, on AWS and Kubernetes. Experienced in compliance-driven delivery: PCI DSS vulnerability remediation, audit logging for client security audits, and hardening services against information leaks.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

APIs and services in a regulated domain

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals (agent, customer, timestamp, session, IP) to meet client security and compliance audits.
- Standardised input validation across partner microservices and removed stack traces and SQL details from error responses, so malformed requests no longer trigger database errors.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so one partner's failure no longer aborts the run.
- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by loading caches once per batch instead of per file.

Full stack features and testing

- Built full stack features across web and mobile, including a secure e-code resend flow with audit-logged resends and multi-bank email routing.
- Led a cross-team initiative for automated API testing: request validation, shared error handling, safe test-data cleanup endpoints, OpenAPI documentation, unit tests and a stabilised GitHub Actions CI pipeline.

Cloud and platform

- Authored the AWS SDK for JavaScript v2-to-v3 migration guide across Lambdas, Fargate services and microservices; planned testing for Node 22 runtime upgrades and remediated critical Snyk-flagged vulnerabilities for PCI DSS.
- Resolved a customer-facing outage in a Kubernetes/Istio ingress setup and corrected the Helm deployment templates.
- Automated scheduled file processing with EventBridge, Lambda and Fargate.

Team

- Mentored and onboarded four new starters; run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost web API using Go, Docker, Kubernetes and Git.

SKILLS

- **Back end:** Node.js (JavaScript), C#/.NET Core, REST APIs, microservices, OpenAPI/Swagger, GraphQL
- **Data:** MongoDB, PostgreSQL, SQL
- **Cloud and delivery:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Docker, Terraform, GitHub Actions
- **Front end:** Angular, HTML/CSS
- **Security and compliance:** PCI DSS remediation, Snyk, OWASP, audit logging

PROJECTS

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- Web app flagging irregular card transactions, built as Docker/Kubernetes microservices on AWS.

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026